

Emran Masoumifeshani, PhD

Sr. AI Data Scientist | AI/ML Engineer

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PROFILE

PhD-trained **AI Engineer** with 7+ years of experience spanning AI engineering, machine learning, statistical modelling, computational science, and data-driven software development. Experienced in designing and implementing **end-to-end AI systems**, from data preparation and modelling to API development, evaluation, containerisation, and production deployment.

Hands-on experience building **Generative AI, RAG, agentic AI, recommendation, computer vision, and machine learning applications**, with strong foundations in statistical learning, predictive modelling, feature engineering, model evaluation, uncertainty analysis, and scientific computing. Strong programming background in **Python and C++**, combining software engineering practices with rigorous quantitative modelling.

Experienced with **LLMs, embeddings, vector retrieval, LangChain, LangGraph, FastAPI, Docker, cloud-based data platforms, SQL/NoSQL, and MLOps workflows**. Brings strong machine learning capability together with an engineering mindset focused on building reliable, scalable, reproducible, and production-oriented AI systems.

CORE AI & TECHNICAL SKILLS

AI Engineering & Architecture: Applied AI, AI System Architecture, End-to-End AI Applications, AI APIs, Intelligent Applications, Production AI Systems, Software Engineering for AI

Generative AI & LLMs: LLMs, Transformers, RAG, Embeddings, Vector Retrieval, LangChain, LangGraph, Prompt Engineering, Tool Calling, Agentic AI, LLM Evaluation

Machine Learning: Predictive Modelling, Statistical Learning, Supervised Learning, Unsupervised Learning, Clustering, Customer Segmentation, Feature Engineering, Feature Extraction, Model Development, Model Evaluation, Model Benchmarking

Computer Vision & Recommendation: Image Processing, Computer Vision, Product Recommendation, Similarity-Based Retrieval, Visual Product Workflows, Personalisation

MLOps & Software Engineering: Python, C++, FastAPI, REST APIs, Docker, Nginx, CI/CD, Git, Linux, Model Versioning, Experiment Tracking, Reproducible Deployment

Data Engineering & Cloud: SQL, Pandas, Spark, PySpark, BigQuery, Databricks, MongoDB, NoSQL, Vector Databases, ETL, Data Modelling, GCP, AWS, Azure

Statistics & Scientific ML: Statistical Modelling, Uncertainty Quantification, Quantitative Analysis, High-Dimensional Data Analysis, Scientific Machine Learning, Numerical Simulation, Computational Modelling, Quantitative Validation

Responsible AI & Evaluation: AI/ML Evaluation, Explainability, SHAP, Bias & Fairness, Model Reliability, Model Lifecycle Governance, GDPR

PROFESSIONAL EXPERIENCE

Founder / AI Engineer
Royal Dywany, Warsaw, Poland

Feb 2025 – Present

- Designed and developed **AI-driven e-commerce systems**, combining AI engineering, machine learning, computer vision, recommendation workflows, and Generative AI.
- Architected an **AI-powered product recommendation and visualisation system** using Python and FastAPI, incorporating image-processing and recommendation components based on product similarity, style, colour, and user preferences.
- Developed modular **AI services and REST APIs** with Python/FastAPI and deployed containerised applications using Docker and Nginx with HTTPS on a cloud server.

- Designed data and retrieval infrastructure covering **ETL, data modelling, SQL/NoSQL databases, vector databases, and cloud-based data platforms.**
- Designed and implemented **RAG and agentic AI workflows** using LangChain and LangGraph, including retrieval architectures, tool-based workflows, and structured LLM evaluation.
- Developed evaluation workflows using **LangSmith and DeepEval** to assess AI-system accuracy, robustness, reliability, explainability, and potential bias.
- Applied software engineering and MLOps practices including **containerisation, version control, API development, CI/CD, reproducible deployment, and model lifecycle management.**
- Established technical practices for responsible AI development, including model lifecycle governance and GDPR-oriented data handling.
- Coordinated a cross-functional team of 5 across AI development, data, product, and e-commerce operations.

Senior Research Assistant & Assistant Professor

Sep 2023 – May 2025

The Institute of Organic Chemistry, Polish Academy of Sciences (IOC PAS), Warsaw

- Conducted advanced computational research involving **large-scale scientific datasets, predictive modelling, statistical analysis, and quantitative model validation.**
- Developed Python-based computational workflows for **data preprocessing, feature engineering, feature extraction, automated data generation, simulation, statistical analysis, and reproducible experimentation.**
- Applied **machine learning and data-driven modelling concepts** to complex scientific problems involving high-dimensional molecular and computational data.
- Designed quantitative evaluation and benchmarking workflows assessing **model accuracy, robustness, uncertainty, reliability, and generalisation.**
- Analysed high-dimensional simulation and experimental datasets to identify quantitative relationships and support scientific prediction and decision-making.
- Developed reproducible computational pipelines and automated research workflows using programming, numerical methods, statistical analysis, and version-controlled software.
- Collaborated across computational and experimental research teams and communicated quantitative results through peer-reviewed scientific publications.

PhD Researcher / Computational Scientist

2018 – 2023

University of Warsaw, Warsaw, Poland

- Developed the **AROFrag computational methodology** for systematic decomposition and quantitative analysis of aromatic molecular systems.
- Designed Python/C++ software components and computational workflows for **molecular data processing, feature extraction, automated data generation, large-scale calculations, and statistical analysis.**
- Developed algorithms for systematic molecular fragmentation and automation of repetitive computational tasks in large-scale quantum-chemistry workflows.
- Performed quantitative model validation and benchmarking against DFT and ab initio calculations, analysing **accuracy, robustness, uncertainty, and generalisation.**
- Applied statistical modelling, uncertainty quantification, quantitative data analysis, and hypothesis-driven evaluation to computational research problems.
- Built reproducible computational workflows and software validated through peer-reviewed scientific research.

R&D Engineer

Nov 2013 – Nov 2018

Nirou Trans, Iran

- Conducted quantitative engineering analysis and computational evaluation of electrical transformer systems and experimental datasets.
- Developed and applied systematic testing, data-analysis, and engineering evaluation methods to support R&D activities.

Selected Machine Learning & AI Projects

- **Customer Segmentation & E-Commerce Analytics** *Python, Pandas, scikit-learn*
 - Built an end-to-end customer analytics and machine learning pipeline from transactional e-commerce data, transforming raw records into customer-level features.
 - Performed data cleaning, transformation, customer-level feature engineering, and RFM-style behavioural analysis.
 - Applied clustering-based customer segmentation and analysed segment purchasing behaviour to connect machine learning outputs with targeting, retention, and marketing decisions.
- **AI-Powered Product Recommendation System** *Python, FastAPI, Computer Vision, Docker, Nginx*

- Designed an AI application for an e-commerce rug business combining image-processing and recommendation components to help customers visualise and select products for their rooms.
- Implemented recommendation workflows based on product similarity, style, colour, and user preferences, together with modular REST API components.
- Deployed the application using Docker and Nginx with HTTPS on a cloud server, providing a production-oriented foundation for further personalised AI recommendations.
- **RAG Document Assistant** *Python, LangChain, RAG, LLMs, FastAPI*
 - Built a modular Retrieval-Augmented Generation pipeline for natural-language question answering over private documents.
 - Implemented document ingestion, chunking, embedding generation, vector-based context retrieval, prompt handling, and grounded LLM generation.
 - Developed modular retrieval and inference components and exposed the application through FastAPI endpoints for integration with external applications.
- **AROFRAG – Scientific Computing & Automation Software** *Python, C++, Algorithms, Scientific Computing*
 - Developed scientific computing software for systematic molecular fragmentation and automation of computational chemistry workflows.
 - Implemented Python and C++ components for molecular-structure processing, algorithm development, automation, and large-scale quantum chemistry calculations.
 - Established reproducible computational workflows that reduced repetitive manual work and enabled systematic validation through peer-reviewed scientific research.

Education

Doctor of Philosophy (Ph.D.) in Chemical Sciences

2018–2023

University of Warsaw, *Warsaw, Poland*

- Dissertation: Developed a novel computational methodology for systematic molecular fragmentation of branched molecules and polycyclic aromatic hydrocarbons, enabling scalable quantitative modelling of large molecular systems.

Master of Science (M.Sc.) in Physical Chemistry

2011–2013

University of Tehran, *Tehran, Iran*

- GPA: **18.66/20**

Ranked 1st in Class

Bachelor of Science (B.Sc.) in Chemistry

2007–2011

University of Tehran, *Tehran, Iran*

- GPA: **17.41/20**

Ranked 1st in Class

Selected Publications

- **E. Masoumifeshani**, T. Korona, “Intermolecular Interaction Energies with AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Computational Chemistry*, 2024, **45**(29), 2446–2458. [doi:10.1002/jcc.27429](https://doi.org/10.1002/jcc.27429)
- **E. Masoumifeshani**, T. Korona, “AROFRAG—A Systematic Approach for Fragmentation of Aromatic Molecules,” *Journal of Chemical Theory and Computation*, 2024, **20**, 1078–1087. [doi:10.1021/acs.jctc.3c00875](https://doi.org/10.1021/acs.jctc.3c00875)
- A. Pareek, M. Morawiak, **E. Masoumifeshani**, P. Gawel, “Photostable π -Expanded Furans via Base-Mediated Quinone Oxygen-Annulation,” *The Journal of Organic Chemistry*, 2026, **91**, 2837–2843. [doi:10.1021/acs.joc.5c02906](https://doi.org/10.1021/acs.joc.5c02906)
- M. Y. Mehboob, **E. Masoumifeshani**, Z. Badri, C. Foroutan-Nejad, “Design of Carbon–Carbon Ylides,” *Angewandte Chemie International Edition*, 2025, **64**, e202507446. [doi:10.1002/anie.202507446](https://doi.org/10.1002/anie.202507446)
- K. Bartkowski, **E. Masoumifeshani**, M. Kotowska, U. Klimczak, B. Furman, C. Foroutan-Nejad, M. Lindner, “One-Pot Transition-Metal-Free Synthesis of π -Extended Bipolar Polyaromatic Hydrocarbons,” *Angewandte Chemie International Edition*, 2025, **64**, e202423282. [doi:10.1002/anie.202423282](https://doi.org/10.1002/anie.202423282)

Certifications

- Building Generative AI-Powered Applications with Python – IBM
- Generative AI Language Modeling with Transformers – IBM
- Gen AI Foundational Models for NLP & Language Understanding – IBM
- Generative AI and LLMs: Architecture and Data Preparation – IBM
- The Structured Query Language (SQL) – University of Colorado Boulder
- Foundations: Data, Data, Everywhere – Google